A Differential-Inclusion Algorithm for Nonsmooth Convex Resource Allocation

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Abstract—This document exists only to exercise the PDF reader of *dsh-paper-reader-studio*: it is a real, two-page, paper-style PDF with a selectable text layer, an outline, and pages wide enough to show the available reading width. It mirrors the structure of a typical control paper—problem statement, a continuous-time differential inclusion, four convergence statements, and a numerical illustration—so that the reader panel can render pages, scroll, zoom, remember the reading position and turn a selection into a question. None of the claims below is a research result; the text is written to look like a paper, not to report one.

Index Terms—differential inclusion, nonsmooth convex optimization, resource allocation, multi-agent networks, demo document, PDF text layer.

I. Introduction

Resource allocation over a multi-agent network asks a group of agents to split a shared budget while each agent keeps its own objective and constraints private. A centralized solution collects every local function at one node; a distributed one lets each agent exchange only a few numbers with its neighbors. The second setting is attractive when the network is large, when communication is expensive, or when the local data cannot leave the agent that owns it.

Classical distributed schemes either project the state onto the feasible set at every step or require the local objectives to be strongly convex. Both requirements are restrictive in practice: projection needs a global description of the constraint set, and strong convexity rules out piecewise-linear costs that appear in dispatch and scheduling. The demo algorithm below avoids both by combining a network Laplacian with a time-varying penalty whose weight grows without bound.

The reader may use this page to check the interactive features of the panel: select a sentence and a floating bar should appear, click an outline entry on the left to jump to the corresponding page, and watch the toolbar report the page and the zoom level. The remainder of the document is filler with realistic typography and mathematics.

$$f^{*} = \min_{x} \ \Sigma_{i=1}^{n} f_i(x_i) \quad \text{subject to} \quad \Sigma_{i=1}^{n} x_i = b, \ \ g_i(x_i) \le 0 \tag{1}$$

Problem (1) couples the agents only through the equality constraint: the budget is shared, while every inequality constraint is private. This structure is what makes a distributed method possible at all, and it is also where most of the difficulty hides, because the shared budget is a global object that no single agent can observe.

II. Problem Formulation

Let the network be described by a connected undirected graph with Laplacian L. Agent i owns a convex, possibly nonsmooth function f_i and a convex constraint map g_i. We assume that Slater's condition holds locally and that the local feasible sets are bounded, so that the KKT system of (1) has a solution with bounded multipliers.

$$J(x) = \Sigma_i \, \Sigma_k \, max\{0, \, g_{ik}(x_i)\} \geq 0 \quad (2)$$

III. Continuous-Time Algorithm

The algorithm is a differential inclusion driven by the subdifferential of the penalized objective, with the Laplacian acting as an implicit multiplier. The penalty weight grows quadratically in time, which is the mechanism behind the finite-time feasibility statement of Theorem 3.

$$\dot{x}(t) \in -(L \otimes I_d) \, (\, \partial f(x(t)) + (t+1)^2 \, \partial J(x(t)) \,) \quad (3)$$

IV. Convergence Statements

Theorem 1. For every initial state on the resource-constraint set, the inclusion admits a unique solution that stays on that set for all time.

Theorem 2. Every solution of (3) is bounded on the whole time axis.

Theorem 3. The inequality-constrained feasible set is reached in finite time and is invariant afterwards.

Theorem 4. Every solution converges to an optimal point of (1).

V. Numerical Illustration

Four agents, two coordinates per agent, and a budget vector b = (5, 20) are used in the demo. The local costs mix an absolute value, a smooth rational term, a sum of two absolute values and a log-sum-exp, so that the aggregate objective is nonsmooth and not strictly convex. The trajectories settle near x_1 = (0.49, 0.49), x_2 = (4, 1), x_3 = (4.9, 12) and x_4 = (0.41, 0.41). These numbers are illustrative only.

References

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